

Name: _____

Date: _____

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.11.03.

1 Classify by Sides

A triangle has side lengths of 5 cm, 5 cm, and 8 cm.

Classify the triangle by its sides.

Isosceles — exactly two sides are equal. _____

2 Classify by Angles

A triangle has angles of 35° , 90° , and 55° .

Classify the triangle by its angles.

Right triangle — one angle is exactly 90° . _____

3 Double Classification

A triangle has angles of 60° , 60° , and 60° , and all sides are 7 cm.

Classify the triangle by both sides and angles.

Equilateral (all sides equal) and acute (all angles less than 90°). ____

4 Can It Exist?

Sophia says she drew a right isosceles triangle. Its two equal angles are each 45° .

Verify that this triangle is possible by checking that all angles sum to 180° .

$45^\circ + 45^\circ + 90^\circ = 180^\circ$. Yes, a right isosceles triangle is possible. ____

5 Reasoning: Kai's Triangle

Kai drew a triangle and said it is both obtuse and equilateral.

Is this possible? Explain using the definition of each triangle type.

No. An equilateral triangle has all angles equal to 60° , so it is always acute. An obtuse triangle must have one angle greater than 90° .